

JAVA PROGRAMMING LABORATORY

Experiment No: 05

Date: 21/09/2026

Title: Demonstration of SortedMap Interface Methods using TreeMap

1. Objective / Aim

To understand the SortedMap interface in Java and demonstrate commonly used SortedMap methods such as firstKey(), lastKey(), headMap(), tailMap(), subMap(), and comparator() using TreeMap.

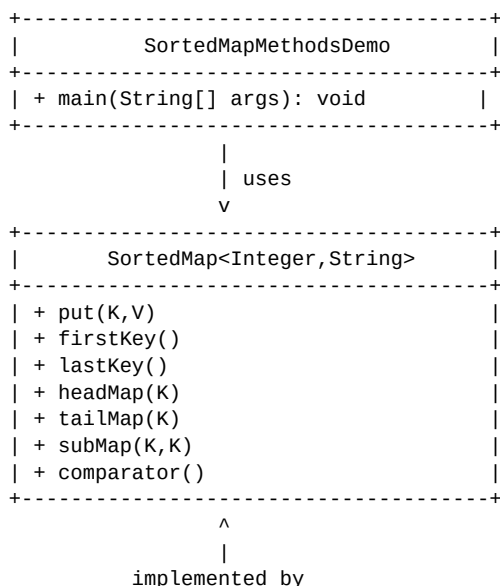
2. System Requirements

Hardware Requirements	Software Requirements
Computer/Laptop	OS: Windows / Linux / macOS
Minimum 4 GB RAM	JDK 17 or above
Minimum 1 GHz processor	Eclipse / IntelliJ IDEA / NetBeans / VS Code
At least 100 MB free storage	Java compiler and command prompt/terminal

3. Algorithm / Logic Description

1. Import SortedMap and TreeMap from java.util.
2. Create a SortedMap using TreeMap.
3. Insert roll numbers and student names using put().
4. Display the map; TreeMap keeps keys in ascending order.
5. Use firstKey() to get the smallest key.
6. Use lastKey() to get the largest key.
7. Use headMap(103) for keys less than 103.
8. Use tailMap(103) for keys greater than or equal to 103.
9. Use subMap(102, 105) for 102 inclusive to 105 exclusive.
10. Use comparator() to check the comparator; natural TreeMap ordering returns null.

4. UML Class Diagram



```

      |
+-----+
|           TreeMap           |
+-----+
| Stores entries in sorted key order |
+-----+

```

5. Source Code

```

import java.util.SortedMap;
import java.util.TreeMap;

class SortedMapMethodsDemo {

    public static void main(String[] args) {

        // Creating a SortedMap using TreeMap
        SortedMap<Integer, String> students = new TreeMap<>();

        // Adding key-value pairs
        students.put(103, "Rohan");
        students.put(101, "Aarav");
        students.put(105, "Diya");
        students.put(102, "Meera");
        students.put(104, "Anjali");

        System.out.println("SortedMap: " + students);
        System.out.println("First Key: " + students.firstKey());
        System.out.println("Last Key: " + students.lastKey());
        System.out.println("HeadMap (< 103): " + students.headMap(103));
        System.out.println("TailMap (>= 103): " + students.tailMap(103));
        System.out.println("SubMap (102 to 105): " + students.subMap(102, 105));
        System.out.println("Comparator: " + students.comparator());
    }
}

```

6. Completion and Execution Commands

```

Step 1: Save as SortedMapMethodsDemo.java
Step 2: Compile: javac SortedMapMethodsDemo.java
Step 3: Execute: java SortedMapMethodsDemo

```

7. Output

```

SortedMap: {101=Aarav, 102=Meera, 103=Rohan, 104=Anjali, 105=Diya}
First Key: 101
Last Key: 105
HeadMap (< 103): {101=Aarav, 102=Meera}
TailMap (>= 103): {103=Rohan, 104=Anjali, 105=Diya}
SubMap (102 to 105): {102=Meera, 103=Rohan, 104=Anjali}
Comparator: null

```

8. Result / Completion

The program was successfully compiled and executed. The SortedMap interface methods were demonstrated using TreeMap, with entries maintained in ascending order of keys.